Inorganic Cume September 13, 1997

Give a brief "descriptive essay" dealing with the metal-ligand chemistry of <u>three</u> of the following ligands. The discussion for the CO ligand below can be used as a style guide

> **<u>CO</u>** - The carbon monoxide ("carbonyl") ligand is primarily a monodentate two electron donor. In bonding to transition metals, CO acts synergistically as a <u>sigma</u> donor and a <u>pi</u> acceptor, or <u>pi</u>-acid ligand. Carbonyl groups are excellent choices to stabilize low oxidation states. The following types of interactions are most common:
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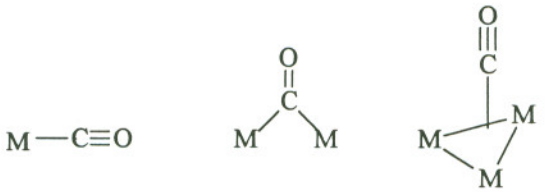
> 
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> The υ(CO) stretching frequency is sensitive to the degree of <u>pi</u> back-donation from the metal and the strength of the M - C <u>sigma</u> bond. The υ(CO) decreases as the extent of back-bonding from the metal increases. Carbonyl ligands are susceptible to nucleophilic attack as in:
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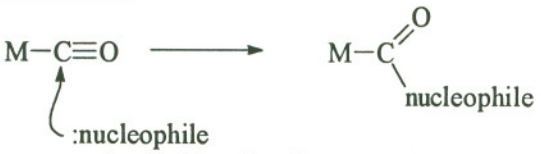
> 
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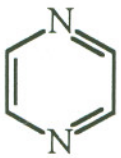
> Metal carbonyls are synthesized by the direct reaction of metal + CO (usually at high T and P) or by reduction of a metal salt in an atmosphere of CO (reductive carbonylation). Metal carbonyls are involved in many photochemical, catalytic and stoichiometric reactions, e.g. olefin carbonylation ……."

Choose <u>three</u> of the following to discuss:

NO	**O_2**	**$(Me_3Si)_3N$**
CH_2CHCH_2	**alkenes**	**NCS^{1-}**
	**C_5H_5**	**H/H_2**
alkoxides		